

KAELEE S. PARKER

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RESEARCH INTERESTS AND EXPERTISE

Observational extragalactic astronomer specializing in the multi-wavelength study of ionizing-photon escape, stellar feedback, and chemical enrichment in low-metallicity star-forming galaxies. My work combines rest-frame UV through mid-infrared spectroscopy from HST, JWST, and ground-based facilities to build empirical frameworks connecting the structure of the interstellar medium to the ionizing output of massive stars. I have extensive experience with spectral modeling, photoionization analysis, IFU data reduction, and statistical comparisons between galaxies in the nearby universe and at high-redshifts. My research bridges nearby resolved systems with the distant galaxies revealed by JWST, with a particular focus on how feedback and gas geometry regulate the escape of ionizing radiation across cosmic time.

Research Keywords: UV/optical/IR spectroscopy • ISM/CGM physics • Ionizing photon escape • Stellar feedback • Low-metallicity galaxies • JWST + HST observations • IFU analysis • Photoionization modeling • Galaxy evolution and reionization

SUMMARY

- **3 first-author papers** (ApJ 2024, 2025; JWST/MIRI paper in prep) and **9 co-authored publications**
- PI of **3 successful observing proposals** (HST Cycle 31, HET UT25-1-011, HET UT21-3-017) and Co-I on two JWST programs
- **Invited talks** at multiple international conferences
- **Mentorship** of several undergraduate researchers
- Awards and fellowships recognizing **research excellence** (Kelton Fellowship 2024, Smith Astronomy Fellowship 2021)
- **Extensive outreach and teaching** record, including course instruction, planetarium education, and leadership in public astronomy programs

EDUCATION

2021 – PRES	University of Texas at Austin ◦ Ph.D. in ASTRONOMY Expected graduation: Spring 2026 Advisor: Danielle A. Berg
2019 – 2021	San Diego State University ◦ M.S. in ASTRONOMY Thesis: <i>On the Relationship Between Interstellar Ca II and Na I Absorption, Dust Attenuation, and Neutral Hydrogen in SDSS-IV/MaNGA</i> Advisor: Kate H. R. Rubin
2015 – 2019	Case Western Reserve University ◦ B.S. in ASTRONOMY, B.A. in PHYSICS Thesis: <i>Adding Low-Mass Galaxies to the Baryonic Tully-Fisher Relation</i> Advisor: Stacy McGaugh

PUBLICATIONS

FIRST-AUTHOR	③ Parker, Berg, Jaskot & Rogers 2026 (submission to ApJ expected Nov 2025) <i>Mapping Ionizing Emission in Nearby Low-Metallicity Galaxies with MIRI/MRS</i> ② Parker, Berg, Chisholm, et al. 2025 (accepted, ApJ) <i>CLASSY XIII: Cutting through the Clouds – Comparing Indirect Tracers of Ionizing Photon Escape</i> ① Parker, Berg, Gazagnes, et al. 2024, ApJ, 977, 104 (arXiv:2410.00236) <i>CLASSY XI: Tracing Neutral Gas Properties using UV Absorption Lines and 21-cm Observations</i>
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PUBLICATIONS (CONT.)

CO-AUTHORED

- ⑨ Trevino, Berg, Gazagnes, **Parker**, et al. (submission to ApJ expected Nov 2025)
Interpreting Si II and C II Absorption Lines from the LzLCS+ using Mock Spectra from a Radiation-Hydrodynamic Simulation
- ⑧ Huberty, Carr, ... **Parker**, et al. 2025 (subm, ApJ)
CLASSY XV: Kinematics and Spatial Distributions of Outflows in Local Highly Star-Forming Galaxies
- ⑦ James, Abril-Melgarejo, ... **Parker**, et al. 2025 (subm, ApJ)
CLASSY XIV: Exploring Multi-Phase Metal Enrichment in High-z Analogues & the Mixing of Nitrogen
- ⑥ Mauerhofer, ... **Parker**, et al. 2025 (subm, A&A)
The SPHINX Public Data Release II: Low-ionisation absorption lines and dust attenuation to predict Lyman continuum escape
- ⑤ Arellano-Córdova, ... **Parker**, et al. 2025, MNRAS (accepted; [arXiv:2507.11658](#))
CLASSY XII: Nitrogen Enrichment Shaped by Gas Density and Feedback
- ④ Rubin, Westfall, ... **Parker**, et al. 2025, ApJ, 981, 31 ([arXiv:2502.09707](#))
SDSS-IV MaStar: Quantification and Abatement of Interstellar Absorption in the Largest Empirical Stellar Spectral Library
- ③ Arellano-Córdova, Berg, ..., **Parker**, et al. 2024, ApJ, 968, 98A ([arXiv:2403.08401](#))
CLASSY IX. The Chemical Evolution of the Ne, S, Cl, and Ar Elements
- ② Hu, Martin, ..., **Parker**, et al. 2023, ApJ, 956, 39H ([arXiv:2307.04911](#))
CLASSY VII. Ly α Profiles: The Structure and Kinematics of Neutral Gas and Implications for LyC Escape in Reionization-era Analogs
- ① McGaugh, Lelli, ..., **Parker**, et al. 2021, AJ, 162, 202 ([arXiv:2109.03251](#))
The Baryonic Tully-Fisher Relation in the Local Group and the Equivalent Circular Velocity of Pressure Supported Dwarfs

Astrobiters Undergraduate Research Article, June 2021

OBSERVING PROGRAMS

PI PROGRAMS

- | | |
|--|----------------------|
| Hubble Space Telescope 17430 ▷ COS | \$75,000 25 orbits |
| ◦ Pulling Back the Curtain Veiling Extreme UV Galaxies: Revealing the Mysterious Sources of He II Emission | |
| Hobby-Eberly Telescope UT25-1-011 | 9 hours |
| ◦ Searching for Intermediate Mass Black Holes in Nearby Extreme Emission Line Galaxies | |
| Hobby-Eberly Telescope UT21-3-017 | 5 hours |
| ◦ Star Cluster Spectroscopy as a Tool for Investigating Galactic Outflows | |

CO-I PROGRAMS

- | | |
|---|----------------------|
| James Webb Space Telescope 7041 ▷ MIRI | \$421,196 79 hours |
| ◦ The CLASSYIR Treasury: Unveiling the Cosmic Engines Powering Galaxies with JWST/MIRI (PI Berg) | |
| James Webb Space Telescope 7081 ▷ NIRSpec | \$188,206 26 hours |
| ◦ A Race Against the Clock: Too Much Nitrogen, Too Early? (PI Berg) | |
| Green Bank Telescope 22B-359 | 34.5 hours |
| ◦ The Role of Gas in the Chemical Evolution of Low-Mass Galaxies (PI McQuinn) | |
| Harlan J. Smith 2.7m Telescope | 5 nights |
| ◦ Mapping the Fundamental Metallicity Relationship of Extremely Metal-poor Galaxies (PI Menchaca) | |

ADDITIONAL OBSERVING EXPERIENCE

- 2024 | **Harlan J. Smith 2.7m Telescope** ▷ [5 nights](#)
◦ Mapping the Fundamental Metallicity Relationship of Extremely Metal-poor Galaxies
- 2023 | **Harlan J. Smith 2.7m Telescope** ▷ [4 nights](#)
◦ Training other UT students
- 2021 | **Harlan J. Smith 2.7m Telescope** ▷ [5 nights](#)
◦ Disentangling the True Nature of Void Galaxies with Secure Oxygen Abundance
Harlan J. Smith 2.7m Telescope ▷ [4 nights](#)
◦ The impact of internal and global properties in chemical abundance determinations in Galactic H II regions
Green Bank 100-m Telescope
◦ Completed training

TALKS, CONFERENCES, AND WORKSHOPS

- 2025 | **Upcoming: Astronomy Colloquium** | Rutgers University
◦ [Invited Talk](#): *TBD*
Feedback with Friends Conference | Ascona, Switzerland
◦ [Invited Talk](#): *Cutting through the Clouds Comparing Indirect Tracers of Ionizing Photon Escape*
IR Fine Structure Line Workshop | Winona
Astronomy Colloquium | University of Minnesota Twin Cities
The Inaugural Cosmic Frontier Center Conference | University of Texas at Austin
Escape of Lyman Radiation Conference | Crete, Greece
Astronomy Colloquium | University of Texas at Austin
- 2024 | **Cosmic Dawn at High Latitudes: Workshop and Conference** | Nordita, Stockholm, Sweden
CFC/CCA Joint Workshop | University of Texas at Austin
Astronomy Colloquium | University of Texas at Austin
- 2023 | **52nd Saas-Fee Advanced Course: The CGM across Cosmic Time** | Les Diablerets, Switzerland
Beyond Astronomy Conference: UV Galaxies | Reykjavik, Iceland
◦ [Invited Talk](#) : *Tracing Photon Escape Through Neutral Gas in CLASSY Galaxies*
2nd Year Research Defense | University of Texas at Austin
◦ *Tracing Photon Escape Through Neutral Gas Around Star-Forming Galaxies*
- 2022 | **XXXIst General Assembly of the International Astronomical Union** | Busan, South Korea
Astronomy Colloquium | University of Texas at Austin
IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution | Granada, Spain
Galaxy Evolution Workshop | virtual
◦ [Invited Talk](#) : *Mapping the Source of Extreme Emission in a Nearby Dwarf Galaxy*
- 2021 | **Astronomy Colloquium** | University of Texas at Austin
M.S. Thesis Defense | San Diego State University
◦ *On the Relationship Between Interstellar CII and NaI Absorption, Dust Attenuation, and Neutral Hydrogen in SDSS-IV/MaNGA*
SAZERAC 2.0 Conference | virtual
237th Meeting of AAS | virtual
- 2020 | **SDSS-IV Month of Online MaNGA** | virtual
GROWTH Astronomy Summer School | San Diego State University
Women in Astronomy | SDSU Planetarium

TALKS, CONFERENCES, AND WORKSHOPS (CONT.)

- 2019 | **Undergraduate Capstone Defense** | Case Western Reserve University
to | **Women in Astronomy** | Shafran Planetarium, Cleveland Museum of Natural History
2017 | ◦ Article: <https://tinyurl.com/5n92nv5h>
| **Conference for Undergraduate Women in Physics** | Toledo, Ohio
| **Dwarf Galaxies on the Shoulders of Giants Conference** | Cleveland, Ohio

AWARDS

Fellowships:

- 2024 | **PHILLIP AND KAREN KELTON FELLOWSHIP:** [3 months of funding](#)
2021 | **RUTH AND CLIFFORD SMITH ASTRONOMY FELLOWSHIP:** [\\$1500 in funding](#)

Scholarships:

- 2020 | **WILLIAM F. LUCAS/SAN DIEGO ASTRONOMY ASSOCIATION ENDOWMENT**
| **REGINALD F. BULLER ENDOWMENT SCHOLARSHIP**
| **AWONA W. HARRINGTON ASTRONOMY SCHOLARSHIP**
| **JAMES AND MARY CROUCH MEMORIAL SCHOLARSHIP**
| **MASTER'S RESEARCH SCHOLARSHIP**
2018 | **JUNIOR/SENIOR SCHOLARSHIP**

MENTORSHIP

- 2025-present | **Alyson Tran** | Undergraduate at UT Austin
 ◦ Summer 2025 research funding award
 ◦ 1st author ApJ paper in progress
2024-present | **John Trevino** | Undergraduate at UT Austin
 ◦ 1st author paper submission to ApJ expected Nov 2025
 ◦ Awarded a 2025 summer NSF REU position at Rochester Institute of Technology
2024 | **C.J. Quintana** | UT Austin REU scholar
 ◦ Presented research poster at the 245th AAS Winter Meeting 2025
2022-2024 | **Alexander Menchaca** | Undergraduate at UT Austin
 ◦ Currently a Ph.D. student in Astronomy at New Mexico State University
 ◦ PI of Harlan J. Smith 2.7m Telescope Program (awarded 5 nights)
 ◦ Awarded 2023 Department of Defense Summer Internship
 ◦ Presented research poster at the 239th AAS Winter Meeting 2023
 ◦ Awarded 2022 Astronomy Department Summer Undergraduate Research Fellowship

TEACHING EXPERIENCE

“Teaching Astronomy” Graduate Course (2023)

Certificate in Evidence-Based Teaching (2020)

Outstanding Supplemental Instruction Leader Award (2019)

- 2019 – 2021 | **Teaching Associate (SDSU)**
2020 – 2021 | **Lead Teaching Associate**
| **Instructor for ASTR 109 - Introductory Astronomy Laboratory.** I created the syllabus, lectures, and assignments related to the labs for this course. As Lead Teaching Associate, I was the bridge between the faculty supervisor and the other TAs. I assigned the lab sections, lead correspondence with the department faculty, and ran the weekly TA meetings. I also helped to revise the course's lab manual, adding and rewriting large sections of it, in order to make it suitable for online teaching.
2019 | **Teaching Assistant (CWRU)**
| **ENGR 131B - Introduction to Engineering and Programming**
| I worked with a team of two other students and one professor to develop this pilot course. My role in its development was to determine the programming content for the course. I taught weekly lectures on programming concepts during the first semester this course was available to students.

TEACHING EXPERIENCE (CONT.)

2017 – 2019	Supplemental Instruction (SI) Leader (CWRU)
2018 – 2019	Supplemental Instruction (SI) Mentor <i>ENGR 131 - Elementary Computer Programming</i> I facilitated review sessions with students in a traditionally difficult course. I created session plans that catered to a variety of learning styles for diverse groups ranging in size from 1-90 students. As an SI Mentor: I organized meetings for mentees to discuss SI concepts, experiences, and provide advice to one another. I also observed and supplied feedback to each mentee on their sessions over the course of the semester, as well as assisted in the SI Leader recruitment and training.

OUTREACH AND SERVICE

Certificate of Distinguished Service, awarded April 2018

2022 – PRESENT	Girl Day <i>University of Texas at Austin</i> I have co-lead the involvement of AWARE (Association of Women in Astronomy Research and Education) in Girl Day for 3 years. This involved organizing activities, volunteers, and performing planetarium shows on the day of the event.
2021 – PRESENT	Affiliate <i>The Starbound Foundation</i> (Website: https://www.thestarboundfoundation.org/) I am affiliate of this charity, which aims to make astronomy more accessible to elementary schools and their students. We bring our donated portable StarLab planetariums to local schools, focusing on those that are under-served and have fewer resources for fieldtrips.
2019 – 2021	Astronomy Public Outreach <i>San Diego State University</i> I began presenting planetarium shows through the SDSU Astronomy Department early on in my first semester. Many of our shows for the community have been for students in under-served school systems around San Diego. When COVID-19 began, I was working with the Public Outreach Coordinator in the department, organizing more virtual outreach events –from unique virtual planetarium shows to a virtual observing night– and working with members of the community, like the Pre-College Institute, on plans for future collaboration.
2017 – 2019	Public Outreach and Planetarium Assistant <i>Cleveland Museum of Natural History</i> I helped with astronomy public outreach programs such as Astronomy Day, Rocket Camp, and Museum Under the Stars. I presented star shows in the Shafran planetarium multiple times a week. In total, I spent over 150 hours volunteering at the Cleveland Museum of Natural History.